

A Proposed Solution to Erdős Problem 788

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Abstract

For $n \geq 1$, put

$$I_n = (n, 2n) \cap \mathbb{N}, \quad J_n = (2n, 4n) \cap \mathbb{N},$$

let $f(n)$ be the largest integer t such that every $B \subseteq J_n$ admits a set $C \subseteq I_n$ with $c + c' \notin B$ for distinct $c, c' \in C$ and $|B| + |C| \geq t$. We prove that there is an absolute constant $c > 0$ such that, for all sufficiently large n ,

$$c\sqrt{n \log n} \leq f(n) \leq n^{\frac{1}{2} + O\left(\left(\frac{\log \log n}{\log n}\right)^{1/3}\right)}.$$

In particular, $f(n) = n^{1/2 + o(1)}$, which gives an affirmative answer to the principal question for every sufficiently large n .

The upper bound is obtained from a family of $p^{o(r)}$ surjective $\mathbb{F}_p$ -linear maps $\mathbb{F}_p^{2r} \rightarrow \mathbb{F}_p^r$ which is a strong seeded extractor at min-entropy $r + o(r)$. Taking the union of their kernels gives a palette of $p^{r+o(r)}$ group sums with independence number $p^{r+o(r)}$. We give a self-contained mixed-alphabet Trevisan–RRV reconstruction argument establishing the extractor, and then lift the construction to ordinary integer sums while accounting for every base- p carry. An ordered block design uses the unused suffix of each hybrid to pay for its overlap excess, which gives the stated quantitative rate. The lower bound follows from the sparse-neighborhood coloring theorem of Alon, Krivelevich, and Sudakov. This proposed solution was found by GPT-5.

1 Introduction

Throughout, $\mathbb{N} = \{1, 2, \dots\}$, and all logarithms are natural unless a base is explicitly indicated. For $n \in \mathbb{N}$, put

$$I_n = (n, 2n) \cap \mathbb{N}, \quad J_n = (2n, 4n) \cap \mathbb{N}.$$

For $B \subseteq J_n$, call $C \subseteq I_n$ *B-admissible* if $c + c' \notin B$ whenever $c, c' \in C$ are distinct, and define

$$f(n) = \max\{t \in \mathbb{Z} : \text{every } B \subseteq J_n \text{ has a } B\text{-admissible } C \text{ with } |B| + |C| \geq t\}.$$

For a finite nonempty set W , write U_W for its uniform distribution. For probability distributions P, Q on the same finite set, write

$$d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|.$$

When Q has full support, also write

$$D(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}.$$

This is Problem 788 in the Erdős Problems collection [1]. The collection cites Erdős's 1973 survey [2], which credits Choi's 1971 paper [3] with raising the problem.

Choi proved $f(n) \ll n^{3/4}$ and conjectured the bound $f(n) \ll_\varepsilon n^{1/2+\varepsilon}$ [3]. Baltz, Schoen, and Srivastav improved the upper bound to $f(n) \ll (n \log n)^{2/3}$ [4]. The online problem record reports the subsequent bound $f(n) \leq n^{3/5+o(1)}$, obtained by combining a random-Cayley-graph reduction with work of Alon and Pham [1, 5]. None of these upper bounds is used below; our construction directly gives the conjectured exponent for every sufficiently large interval length.

Our main theorem gives a little more quantitative information than the requested exponent.

Theorem 1.1 (Main theorem). *There are absolute constants $c, C > 0$ such that, for all sufficiently large n ,*

$$c\sqrt{n \log n} \leq f(n) \leq n^{\frac{1}{2}+C\left(\frac{\log \log n}{\log n}\right)^{1/3}}.$$

Consequently

$$f(n) = n^{1/2+o(1)}.$$

Remark 1.2 (Quantifiers in the upper bound). The theorem supplies a witness palette for every sufficiently large integer n , not merely along a subsequence. Equivalently, for each fixed $\varepsilon > 0$ and all sufficiently large n , one can choose a single $B \subseteq J_n$ such that every B -admissible $C \subseteq I_n$ obeys $|B| + |C| \leq n^{1/2+\varepsilon}$.

The two sides of the proof use different information. The lower bound holds for every palette B . For the upper bound we construct one palette, depending on n , by combining coding theory, weak designs, and an exact carry closure.

The proof is organized as follows. Section 2 converts the problem into an exact minimization involving the independence number of a sum graph. Section 3 proves the universal lower bound. Section 4 constructs and verifies a mixed-alphabet linear seeded extractor, using an ordered weak design whose overlap excess is paid for by the remaining hybrid coordinates; Section 5 converts its kernels into a finite-field sum palette. Section 6 lifts that palette to ordinary integer sums without discarding any carry pattern. Section 7 chooses the parameters uniformly for every interval length, and Section 8 translates the construction back to the original intervals and completes the proof.

2 Exact graph formulation

For $s \in J_n$, let M_s be the graph on I_n whose edges are the unordered pairs $\{c, c'\}$ satisfying

$$c \neq c', \quad c + c' = s.$$

Each M_s is a matching: after one endpoint is specified, the other is forced. A possible equality $c = s - c$ creates no loop, since the problem only constrains distinct elements. For $B \subseteq J_n$, put

$$G_B = \bigcup_{s \in B} M_s.$$

Proposition 2.1 (Min–max identity). *For every $n \geq 1$,*

$$f(n) = \min_{B \subseteq J_n} (|B| + \alpha(G_B)).$$

Moreover, the minimum is unchanged if B is required to consist only of sums of two distinct elements of I_n .

Proof. A set $C \subseteq I_n$ is B -admissible exactly when it is independent in G_B . Thus, for fixed B , the largest possible value of $|B| + |C|$ is $|B| + \alpha(G_B)$. An integer t has the universal property in the definition of $f(n)$ exactly when

$$t \leq |B| + \alpha(G_B) \quad \text{for every } B \subseteq J_n.$$

The family of choices of B is finite, so taking the largest such t gives the displayed minimum.

If s is not the sum of two distinct elements of I_n , then M_s is empty. Deleting such an s from B leaves G_B unchanged and decreases $|B|$. Hence no minimizer needs an unattainable sum. $\square$

It will be useful to normalize the vertices. Put

$$N = n + 1, \quad [N]_0 = \{0, 1, \dots, N - 1\}.$$

Write $c = n + 1 + x$. Then

$$c + c' = 2n + 2 + x + y.$$

For $N \geq 2$, the sums of two distinct members of $[N]_0$ are exactly

$$\{1, 2, \dots, 2N - 3\}.$$

For $A \subseteq \{1, \dots, 2N - 3\}$, let H_A be the graph on $[N]_0$ in which distinct x, y are adjacent exactly when $x + y \in A$. Translation identifies H_A with G_B for

$$B = 2n + 2 + A \subseteq \{2n + 3, \dots, 4n - 3\} \subseteq J_n.$$

We will construct and estimate H_A and translate back only at the end. If $b = |A|$, then

$$\Delta(H_A) \leq b, \tag{2.1}$$

because each selected sum supplies at most one neighbor of any fixed vertex.

3 Universal lower bounds

We first record an elementary bound which makes the square-root scale visible.

Lemma 3.1 (Edge sums control chromatic number). *Let a simple graph have distinct real labels on its vertices, and suppose that its edges use b distinct sums of endpoint labels. Then*

$$\chi(G) \leq \left\lfloor \frac{b + 3}{2} \right\rfloor.$$

Proof. Put $k = \chi(G)$ and pass to a vertex-minimal induced subgraph K with chromatic number k . Then $\delta(K) \geq k - 1$. Let u and v be the least and greatest labels in K . The at least $k - 1$ sums on edges incident with u are distinct and at most $u + v$. The at least $k - 1$ sums on edges incident with v are distinct and at least $u + v$. The two collections intersect in at most the one value $u + v$. Hence $b \geq 2k - 3$, which is equivalent to the assertion. $\square$

Taking the largest color class in H_A , where $b = |A|$, gives

$$\alpha(H_A) \geq \left\lceil \frac{N}{\lfloor (b + 3)/2 \rfloor} \right\rceil.$$

Optimizing this elementary expression yields, for $N \geq 4$,

$$b + \alpha(H_A) \geq \lceil \sqrt{8N} \rceil - 3.$$

The case $b = 0$ gives the value N and cannot minimize when $N \geq 4$ (the choice $A = \{N - 1\}$ has $b = 1$ and gives $1 + \lceil N/2 \rceil \leq N$). For $b \geq 1$ the relevant values are $b = 2q - 3$, and

$$\min_{q \geq 2} \left(2q - 3 + \left\lceil \frac{N}{q} \right\rceil \right) = \lceil \sqrt{8N} \rceil - 3.$$

For completeness, put $t = \lceil \sqrt{8N} \rceil$. The inequality $2q + N/q \geq \sqrt{8N}$ gives the lower direction. Conversely,

$$\max_{q \in \mathbb{Z}} q(t - 2q) = \left\lfloor \frac{t^2}{8} \right\rfloor \geq N.$$

For $N \geq 4$ a maximizing positive integer q can be chosen with $q \geq 2$; then $N/q \leq t - 2q$ and hence $2q + \lceil N/q \rceil \leq t$. This proves the identity. This already proves $f(n) \geq 2\sqrt{2n} - O(1)$. We next gain a logarithmic factor.

Lemma 3.2 (Triangle sum triples). *If H_A uses b selected sums and has T triangles, then*

$$T \leq \binom{b}{3}.$$

Proof. A triangle with vertices x, y, z has the three pairwise distinct edge sums

$$x + y, \quad x + z, \quad y + z.$$

Their unordered set determines the triangle: if the three incident sums are denoted by s_{xy}, s_{xz}, s_{yz} , then

$$x = \frac{s_{xy} + s_{xz} - s_{yz}}{2}, \quad y = \frac{s_{xy} + s_{yz} - s_{xz}}{2}, \quad z = \frac{s_{xz} + s_{yz} - s_{xy}}{2}.$$

Equivalently, from an unordered triple $\{a, b, c\}$ one recovers the unordered vertex set

$$\left\{ \frac{a + b - c}{2}, \frac{a + c - b}{2}, \frac{b + c - a}{2} \right\}.$$

Thus triangles inject into the three-element subsets of A . □

We use the following conservative form of a theorem of Alon, Krivelevich, and Sudakov [6, Theorem 1.1].

Theorem 3.3 (Sparse-neighborhood coloring theorem, safe form). *There is an absolute $K > 0$ with the following property. If a graph has maximum degree at most d and every vertex neighborhood spans at most d^2/F edges, where $2 \leq F \leq d^2$, then*

$$\chi(G) \leq K \frac{d}{\log F}.$$

The closed endpoint $F = d^2$ follows from the usual strict-range formulation by applying it with $F' = d^2/2$ and changing the absolute constant; bounded d is absorbed by greedy coloring. The primary statement calls d the maximum degree. If that convention is read as requiring equality rather than an upper bound, one may adjoin a disjoint star $K_{1,d}$: this makes the maximum degree exactly d , adds no edges to any neighborhood, and does not affect the coloring bound for the original component.

Proposition 3.4 (Universal lower bound). *There is an absolute $c > 0$ such that, for every sufficiently large N and every $A \subseteq \{1, \dots, 2N - 3\}$,*

$$|A| + \alpha(H_A) \geq c\sqrt{N \log N}.$$

Consequently $f(n) \gg \sqrt{n \log n}$.

Proof. Write $b = |A|$. For a vertex v , let t_v be the number of edges spanned by its neighborhood, equivalently the number of triangles containing v . Lemma 3.2 gives

$$\sum_v t_v = 3T \leq 3 \binom{b}{3} \leq \frac{b^3}{2}.$$

Therefore the set

$$U = \left\{ v : t_v \leq \frac{b^3}{N} \right\}$$

has at least $N/2$ vertices. The induced graph $H = H_A[U]$ has maximum degree at most b , and every neighborhood in H spans at most b^3/N edges.

Suppose $2 \leq b \leq N/2$, and put

$$F_0 = \min \left\{ b^2, \frac{N}{b} \right\} \geq 2.$$

In either case $b^3/N \leq b^2/F_0$. Applying Theorem 3.3 to H and taking a largest color class gives

$$\alpha(H_A) \geq c_0 \frac{N}{b} \log \left(\min \left\{ b^2, \frac{N}{b} \right\} \right) \quad (3.1)$$

for an absolute $c_0 > 0$.

Now split into three ranges. If $b \geq \sqrt{N \log N}$, the palette itself proves the result. If $b < N^{1/4}$, the maximum-degree bound (2.1), followed by greedy coloring, gives

$$\alpha(H_A) \geq \frac{N}{b+1} \gg \sqrt{N \log N}.$$

In the remaining range

$$N^{1/4} \leq b < \sqrt{N \log N},$$

we have

$$\min \left\{ b^2, \frac{N}{b} \right\} \geq \sqrt{\frac{N}{\log N}},$$

and hence its logarithm is at least $(1/3) \log N$ for large N . Equation (3.1) now yields

$$b + \alpha(H_A) \geq b + c_1 \frac{N \log N}{b} \geq 2\sqrt{c_1 N \log N}.$$

The cases $b = 0, 1$ are covered by the maximum-degree estimate. The claim for f follows from Proposition 2.1. $\square$

4 A linear seeded extractor over a growing odd field

The upper bound rests on the following finite-field statement. All logarithms used in min-entropy in this section are to base p .

Theorem 4.1 (Linear extractor family). *Let p run through odd primes and let $r \rightarrow \infty$ in such a way that*

$$(\log(pr))^2 \log(2 + \log(pr)) = o(r \log p).$$

Then there are integers $d, s = o(r)$ and an indexed family

$$\{F_y : \mathbb{F}_p^{2r} \rightarrow \mathbb{F}_p^r : y \in \mathcal{Y}\}, \quad |\mathcal{Y}| \leq p^d,$$

such that every F_y is surjective and $\mathbb{F}_p$ -linear, and every random variable X on $\mathbb{F}_p^{2r}$ with

$$H_{\infty,p}(X) := -\log_p \max_x \Pr[X = x] \geq r + s$$

satisfies

$$\frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} d_{\text{TV}}(F_y(X), U_{\mathbb{F}_p^r}) < \frac{1}{3}.$$

More quantitatively, there is an auxiliary integer $\ell = O(\log(pr))$ for which one may take

$$d, s = O\left(\frac{\ell\sqrt{r} + \ell^2 \log(\ell + 1)}{\log p} + \log_p(pr) + 1\right).$$

The family is indexed: repeated maps are allowed.

The construction is a mixed-alphabet adaptation of the linear form of Trevisan's extractor [8] and the weak-design reconstruction analysis of Raz, Reingold, and Vadhan [7]. The code and every fixed-seed output map are $\mathbb{F}_p$ -linear, but code coordinates and seed coordinates are indexed by binary strings. This decoupling is useful: an overlap of t seed bits has 2^t possible inputs, independently of the growing output alphabet. We prove the required code, design, and reconstruction statements with this distinction explicit.

The proof proceeds in four stages. We first construct a short linear code, and then a binary weak design controlling its overlap moments. A hybrid argument and reconstruction count combine these objects into a strong linear extractor. Finally we choose the parameters and discard rank-deficient seeds; Section 5 turns the retained maps into a sum palette by taking the union of their kernels.

4.1 A short linear list-decodable code

We identify a word of length 2^ℓ with a function $\{0, 1\}^\ell \rightarrow \mathbb{F}_p$. Agreement always means the fraction of coordinates on which two words have the same symbol.

Lemma 4.2 (Short binary-coordinate linear code). *Let p be a prime, $m \geq 1$, and $0 < \eta < 1/2$. There exist a positive integer ℓ and an injective $\mathbb{F}_p$ -linear map*

$$\mathcal{E} : \mathbb{F}_p^m \longrightarrow \mathbb{F}_p^{\{0,1\}^\ell}$$

such that, for every word $Q : \{0, 1\}^\ell \rightarrow \mathbb{F}_p$,

$$\#\left\{x \in \mathbb{F}_p^m : \Pr_{z \sim U_{\{0,1\}^\ell}} [\mathcal{E}(x)(z) = Q(z)] > \frac{1}{p} + \eta\right\} \leq J := \frac{2}{\eta^2} + 1.$$

One can arrange

$$2^\ell \leq \frac{C_0 m \log p}{\eta^4} \tag{4.1}$$

for an absolute constant C_0 .

Proof. Set

$$\tau = \frac{p\eta^2}{2(p-1)}, \quad \Delta = 1 - \frac{1}{p} - \tau.$$

We first find a linear code of relative distance at least Δ . Write

$$H_p(a) = a \log_p(p-1) - a \log_p a - (1-a) \log_p(1-a)$$

for the p -ary entropy function, with the usual continuous extension at the endpoints. Then

$$1 - H_p(\Delta) = \frac{D(P \| U_{\mathbb{F}_p})}{\log p},$$

where

$$P = \left(\frac{1}{p} + \tau, \frac{1}{p} - \frac{\tau}{p-1}, \dots, \frac{1}{p} - \frac{\tau}{p-1}\right).$$

The total variation distance between P and $U_{\mathbb{F}_p}$ is τ . Pinsker's inequality, with natural logarithms [9], gives

$$1 - H_p(\Delta) \geq \frac{2\tau^2}{\log p}.$$

Let $R = 2^\ell$ be the least power of 2 strictly exceeding

$$\frac{(m+1)\log p}{2\tau^2}.$$

The displayed quantity exceeds 1, so $\ell \geq 1$. Choose an $R \times m$ matrix T uniformly over $\mathbb{F}_p$. For every fixed nonzero x , the word Tx is uniform on $\mathbb{F}_p^R$. The standard p -ary Hamming-ball estimate [9] is exact in the form

$$\sum_{0 \leq j \leq aR} \binom{R}{j} (p-1)^j \leq p^{RH_p(a)} \quad \left(0 < a \leq 1 - \frac{1}{p}\right).$$

Indeed, with $z = a/((p-1)(1-a)) \leq 1$, the left side is at most $z^{-aR}(1+(p-1)z)^R = p^{RH_p(a)}$. Taking $a = \Delta$ and using the preceding Pinsker bound show that

$$\Pr[\text{wt}(Tx) < \Delta R] \leq p^{-R(1-H_p(\Delta))}.$$

A union bound over fewer than p^m nonzero x is strictly below one. We may therefore fix T for which every nonzero codeword has relative weight at least Δ . This also makes T injective. Index the R coordinates by $\{0, 1\}^\ell$ and call the resulting encoder $\mathcal{E}$. The choice of R , the formula for τ , and rounding to the next power of 2 give, explicitly,

$$R \leq 2 \frac{(m+1)\log p}{2\tau^2} \leq \frac{8m\log p}{\eta^4},$$

which proves (4.1).

It remains to prove the list bound. Represent the p symbols by unit vectors v_a forming a regular simplex:

$$\langle v_a, v_b \rangle = \begin{cases} 1, & a = b, \\ -1/(p-1), & a \neq b. \end{cases}$$

Represent a word by the normalized concatenation of its simplex vectors. Two words with agreement fraction a then have inner product

$$\frac{pa - 1}{p - 1}.$$

Suppose L codewords agree with Q on more than $1/p + \eta$ of the coordinates. Let their unit vectors be $u_1, \dots, u_L$, and let w be the vector of Q . By this identity,

$$\langle u_i, w \rangle > \gamma := \frac{p\eta}{p-1}.$$

The minimum distance gives, for $i \neq j$,

$$\langle u_i, u_j \rangle \leq \beta := \frac{p\tau}{p-1} = \frac{\gamma^2}{2}.$$

Cauchy–Schwarz now implies

$$L^2\gamma^2 < \left\langle \sum_i u_i, w \right\rangle^2 \leq \left\| \sum_i u_i \right\|^2 \leq L + L(L-1)\beta.$$

Thus

$$L \leq \frac{1-\beta}{\gamma^2-\beta} \leq \frac{2}{\gamma^2} \leq \frac{2}{\eta^2},$$

as required. □

4.2 An ordered binary weak design

Lemma 4.3 (Suffix-slack design). *For all integers $\ell, r \geq 1$, there are an integer D and ℓ -subsets $S_1, \dots, S_r \subseteq [D]$ such that, for every i ,*

$$e_i := \sum_{j < i} \left(2^{|S_i \cap S_j|} - 1 \right) \leq r - i, \quad (4.2)$$

and

$$D = O\left(\ell\sqrt{r} + \ell^2 \log(\ell + 1)\right).$$

The implied constant is absolute, and the sets and their ordering may be chosen deterministically from (r, ℓ) .

Proof. Partition the r rows into consecutive blocks. If R_b rows remain before block b , put

$$m_b = \left\lceil \frac{R_b}{2} \right\rceil, \quad R_{b+1} = \left\lfloor \frac{R_b}{2} \right\rfloor.$$

Thus $R_{b+1} \geq m_b - 1$. Give different blocks disjoint seed-coordinate universes.

For a block of size $m = m_b$, let q be the least prime at least

$$Q = \max\{2, \ell, \lceil \sqrt{m} \rceil\}.$$

Bertrand's postulate gives $q < 2Q$ [10]. Choose an ℓ -set $X \subseteq \mathbb{F}_q$ and use the universe $X \times \mathbb{F}_q$. There are $q^2 \geq m$ affine functions $x \mapsto ax + b$; select any m distinct ones and associate to (a, b) its graph

$$S_{a,b} = \{(x, ax + b) : x \in X\}.$$

Two distinct affine graphs meet in at most one point.

Let a row have zero-based local index t in its block and one-based global index i . Rows in earlier blocks have disjoint universes and contribute zero to the excess in (4.2). Each of the t earlier block-mates contributes at most $2^1 - 1 = 1$. Moreover, the exact number of global rows after this row is

$$r - i = (m_b - t - 1) + R_{b+1}.$$

Consequently

$$e_i \leq t \leq m_b - 1 \leq R_{b+1} \leq r - i.$$

This includes the final global row, for which the last block has size one and the excess is zero.

The universe of block b has size

$$\ell q = O(\ell \max\{\ell, \sqrt{m_b}\}).$$

The block sizes decrease geometrically, so $\sum_b \sqrt{m_b} = O(\sqrt{r})$. Once $m_b < \ell^2$, only $O(\log(\ell + 1))$ nonempty blocks remain, each using $O(\ell^2)$ coordinates. The disjoint union of the block universes therefore has the claimed size; identify it with $[D]$. Taking the least eligible prime and the first affine functions in lexicographic order makes the construction deterministic. $\square$

The subtracted 1 in (4.2) isolates the unavoidable baseline: even disjoint sets contribute $2^0 = 1$ to the reconstruction count. Equivalently,

$$\sum_{j < i} 2^{|S_i \cap S_j|} \leq r - 1.$$

The remaining $r - i$ output coordinates will pay for the overlap excess at hybrid row i .

4.3 The reconstruction argument

Fix

$$\varepsilon = \frac{1}{20}, \quad h = \frac{\varepsilon}{r(p-1)}, \quad \eta = \frac{h}{2}. \quad (4.3)$$

Apply Lemma 4.2 with input dimension $2r$, and let $\mathcal{E}$ be the resulting encoder, with coordinate set $\{0, 1\}^\ell$ and list size $J = 2/\eta^2 + 1$. Apply Lemma 4.3 with these values of ℓ and r . Put

$$d = \left\lceil D \log_p 2 \right\rceil,$$

so that the binary seed set has size $2^D \leq p^d$. For $x \in \mathbb{F}_p^{2r}$ and $y \in \{0, 1\}^D$, define

$$\text{Ext}(x, y)_i = \mathcal{E}(x)(y|_{S_i}), \quad 1 \leq i \leq r. \quad (4.4)$$

The coordinates of each S_i are read in increasing order.

Proposition 4.4 (Extractor inequality). *Suppose the integer s satisfies*

$$s \geq d + \left\lceil \log_p \frac{4J}{h} \right\rceil + 2, \quad r + s \leq 2r. \quad (4.5)$$

If X is any random variable on $\mathbb{F}_p^{2r}$ with $H_{\infty,p}(X) \geq r + s$, and Y is uniform on $\{0, 1\}^D$ independently of X , then

$$d_{\text{TV}}((Y, \text{Ext}(X, Y)), (Y, U_{\mathbb{F}_p^r})) \leq \varepsilon. \quad (4.6)$$

Here the copy of $U_{\mathbb{F}_p^r}$ on the right is independent of Y . For each fixed seed y , the map $x \mapsto \text{Ext}(x, y)$ is $\mathbb{F}_p$ -linear with codomain $\mathbb{F}_p^r$.

Proof. Write $Z = \text{Ext}(X, Y) = (Z_1, \dots, Z_r)$ and suppose that (4.6) fails. Define the hybrids

$$H_i = (Y, Z_1, \dots, Z_i, U_{i+1}, \dots, U_r), \quad 0 \leq i \leq r,$$

where the displayed U_j are independent uniform symbols, also independent of (X, Y, Z) . The triangle inequality gives an i such that

$$\mathbb{E}_{(Y, Z_{<i})} d_{\text{TV}}(\mathcal{L}(Z_i | Y, Z_{<i}), U_{\mathbb{F}_p}) > \frac{\varepsilon}{r}.$$

For a distribution μ on $\mathbb{F}_p$,

$$\max_a \mu(a) - \frac{1}{p} \geq \frac{d_{\text{TV}}(\mu, U_{\mathbb{F}_p})}{p-1},$$

because the total positive deviation is the total variation distance and has at most $p-1$ positive summands. At every context choose a most likely symbol. The preceding inequality produces a deterministic predictor

$$\mathcal{P} : \{0, 1\}^D \times \mathbb{F}_p^{i-1} \rightarrow \mathbb{F}_p$$

such that

$$\Pr[\mathcal{P}(Y, Z_{<i}) = Z_i] > \frac{1}{p} + h.$$

The index i and the predictor $\mathcal{P}$ may depend on the fixed law of X , but from now on they are fixed and do not depend on the point x .

For fixed x , let q_x be the success probability in the preceding predictor bound over Y , with $Z = \text{Ext}(x, Y)$, and put

$$\mathcal{B} = \left\{ x : q_x > \frac{1}{p} + \frac{h}{2} \right\}.$$

Since $q_x \leq 1$, averaging that predictor bound gives

$$\Pr[X \in \mathcal{B}] > \frac{h}{2}. \quad (4.7)$$

Indeed, if $\theta = \Pr[X \in \mathcal{B}]$, then

$$\frac{1}{p} + h < \mathbb{E}q_X \leq \frac{1}{p} + \frac{h}{2} + \theta \left(1 - \frac{1}{p} - \frac{h}{2}\right),$$

which implies $\theta > h/2$.

We now count $\mathcal{B}$. Fix $x \in \mathcal{B}$. Regard a binary assignment on S_i as a code coordinate in $\{0, 1\}^\ell$, using the increasing order on S_i . By averaging over the seed coordinates outside S_i , we obtain a fixing $a \in \{0, 1\}^{[D] \setminus S_i}$ for which the predictor agrees with the word $\mathcal{E}(x)$ on more than a $1/p + h/2 = 1/p + \eta$ fraction of assignments $z \in \{0, 1\}^{S_i}$.

After this outside fixing, the earlier output $\mathcal{E}(x)(y|_{S_j})$ is a function of only the $|S_i \cap S_j|$ overlap coordinates of z . We may enumerate every such function: a function $\{0, 1\}^t \rightarrow \mathbb{F}_p$ has p^{2^t} possibilities. Thus the number of descriptions consisting of the outside fixing and all earlier-output functions is at most

$$\begin{aligned} 2^{D-\ell} \prod_{j < i} p^{2^{|S_i \cap S_j|}} &\leq p^{d + \sum_{j < i} 2^{|S_i \cap S_j|}} \\ &= p^{d + (i-1) + e_i} \leq p^{d+r-1}. \end{aligned}$$

For a description $\mathcal{D} = (a, (g_j)_{j < i})$ and an assignment $z \in \{0, 1\}^{S_i}$, let $y(a, z)$ be the unique seed whose restrictions to S_i and $[D] \setminus S_i$ are respectively z and a , and define

$$Q_{\mathcal{D}}(z) = \mathcal{P}(y(a, z), (g_j(z|_{S_i \cap S_j}))_{j < i}).$$

Via the increasing-order identification $\{0, 1\}^{S_i} \cong \{0, 1\}^\ell$, this is one received word $Q_{\mathcal{D}} : \{0, 1\}^\ell \rightarrow \mathbb{F}_p$. For each $x \in \mathcal{B}$, the description obtained above gives a word $Q_{\mathcal{D}}$ which agrees with $\mathcal{E}(x)$ on more than a $1/p + \eta$ fraction of coordinates. Thus x belongs to the agreement list of at least one counted description. Lemma 4.2 bounds every such list by J , so

$$|\mathcal{B}| \leq Jp^{d+r-1}.$$

The min-entropy assumption now gives

$$\begin{aligned} \Pr[X \in \mathcal{B}] &\leq p^{-r-s} |\mathcal{B}| \\ &\leq Jp^{d-s-1} < \frac{h}{4}, \end{aligned}$$

where (4.5) gives explicitly

$$Jp^{d-s-1} \leq Jp^{-\lceil \log_p(4J/h) \rceil - 3} \leq \frac{h}{4p^3} < \frac{h}{4}.$$

This contradicts (4.7), proving (4.6).

Finally, for fixed y , every coordinate in (4.4) is one coordinate of the $\mathbb{F}_p$ -linear encoding $\mathcal{E}$. The tuple of the r selected coordinates is therefore $\mathbb{F}_p$ -linear. Surjectivity is not asserted yet; it is enforced below. $\square$

4.4 Parameters and surjectivity

From Lemma 4.2, (4.3), and the input dimension $2r$,

$$2^\ell = O(r \log p (pr)^4).$$

Therefore

$$\ell = O(\log(pr)). \quad (4.8)$$

Lemma 4.3 gives

$$D = O\left(\ell\sqrt{r} + \ell^2 \log(\ell + 1)\right)$$

and therefore

$$d = O\left(\frac{\ell\sqrt{r} + \ell^2 \log(\ell + 1)}{\log p} + 1\right).$$

Also

$$\log_p(4J/h) = O(1 + \log_p(pr)),$$

so (4.5) holds with

$$s = O\left(\frac{\ell\sqrt{r} + \ell^2 \log(\ell + 1)}{\log p} + \log_p(pr) + 1\right).$$

The growth hypothesis in Theorem 4.1, together with (4.8), makes both d and s equal to $o(r)$. Indeed, if $u = \log(pr)$ and $\lambda = \log p$, then

$$\left(\frac{u}{\lambda\sqrt{r}}\right)^2 = \frac{u^2}{r\lambda^2} = o(1), \quad \frac{u^2 \log(2+u)}{r\lambda} = o(1),$$

while $\log_p(pr)/r = o(1)$. These three estimates control respectively the $\ell\sqrt{r}$, $\ell^2 \log(\ell + 1)$, and remaining terms above. For the first estimate, use the second one together with $\lambda \geq \log 3$ and $\log(2+u) > 1$. This verifies all parameter claims in Theorem 4.1, except surjectivity.

The full set of 2^D binary seeds is an indexed family of linear maps. Apply Proposition 4.4 to the source $U_{\mathbb{F}_p^{2r}}$. For any source X , the joint distance in (4.6) is exactly the fixed-seed average

$$d_{\text{TV}}((Y, \text{Ext}(X, Y)), (Y, U_{\mathbb{F}_p^r})) = \frac{1}{2^D} \sum_{y \in \{0,1\}^D} d_{\text{TV}}(\text{Ext}(X, y), U_{\mathbb{F}_p^r}).$$

If a seed map has rank $t < r$, its output is uniform on a p^t -element subspace, and therefore

$$d_{\text{TV}}(\text{Ext}(U_{\mathbb{F}_p^{2r}}, y), U_{\mathbb{F}_p^r}) = 1 - p^{t-r} \geq 1 - \frac{1}{p}.$$

If b is the fraction of rank-deficient seeds, (4.6) gives

$$b \leq \frac{\varepsilon}{1 - 1/p} \leq \frac{3}{40}.$$

Retain the good seeds, with their original multiplicities, and put the uniform distribution on this retained index set $\mathcal{Y}$. For each $y \in \mathcal{Y}$, set

$$F_y(x) = \text{Ext}(x, y).$$

Then $|\mathcal{Y}| \leq p^d$, every F_y is surjective, and for every source covered by Proposition 4.4,

$$\frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} d_{\text{TV}}(F_y(X), U_{\mathbb{F}_p^r}) \leq \frac{\varepsilon}{37/40} = \frac{2}{37} < \frac{1}{3}.$$

This completes the proof of Theorem 4.1.

5 From the extractor to a finite-field sum palette

Put

$$G = \mathbb{F}_p^{2r}, \quad V = \mathbb{F}_p^r,$$

and let the maps $F_y : G \rightarrow V$ be those of Theorem 4.1. Define

$$S = \bigcup_{y \in \mathcal{Y}} \ker F_y \subseteq G.$$

Let Γ_S be the simple graph on G in which distinct x, x' are adjacent exactly when $x + x' \in S$.

Proposition 5.1 (Group palette). *With $d, s = o(r)$ as above,*

$$|S| \leq p^{r+d}, \quad \alpha(\Gamma_S) < p^{r+s}.$$

Proof. Every retained map is surjective, so every kernel has size p^r . The definition of S therefore gives

$$|S| \leq |\mathcal{Y}| p^r \leq p^{r+d}.$$

Suppose that $A \subseteq G$ is independent. Fix $y \in \mathcal{Y}$. If $z \neq 0$ and both z and $-z$ occur in $F_y(A)$, choose $a, a' \in A$ with $F_y(a) = z$ and $F_y(a') = -z$. Since p is odd, $z \neq -z$, so $a \neq a'$; but then $a + a' \in \ker F_y \subseteq S$, a contradiction. The zero fibre contains at most one member of A , because two distinct members of that fibre would also have their sum in $\ker F_y$. The nonzero elements of V form $(p^r - 1)/2$ antipodal pairs. Hence

$$|F_y(A)| \leq \frac{p^r + 1}{2}.$$

Any distribution P supported on $T \subseteq V$ satisfies

$$d_{\text{TV}}(P, U_V) \geq 1 - \frac{|T|}{|V|},$$

by testing the event T . If $|A| \geq p^{r+s}$ and X is uniform on A , then $H_{\infty, p}(X) \geq r + s$, while the preceding support bound gives, for every retained seed,

$$d_{\text{TV}}(F_y(X), U_V) \geq \frac{p^r - 1}{2p^r} \geq \frac{1}{3}.$$

This contradicts Theorem 4.1. Therefore $|A| < p^{r+s}$. $\square$

The zero-fibre argument is where the absence of loops must be handled carefully: it forbids two distinct zero-fibre points, but it does not forbid a single point a merely because $2a \in S$.

6 Carry closure and ordinary integer sums

Set

$$M = p^{2r}$$

and identify $[M]_0$ with $\mathbb{F}_p^{2r}$ by the base- p digit bijection

$$\psi(x) = (x_0, \dots, x_{2r-1}), \quad x = \sum_{i=0}^{2r-1} x_i p^i.$$

For $u \in \mathbb{F}_p^{2r}$, define its ordinary carry fibre by

$$\mathcal{C}(u) = \{x + x' : 0 \leq x, x' < M, \psi(x) + \psi(x') = u\}, \quad (6.1)$$

where addition inside $\psi(x) + \psi(x')$ is coordinatewise modulo p .

Lemma 6.1 (Carry count). *For every $u \in \mathbb{F}_p^{2r}$,*

$$|\mathcal{C}(u)| \leq 2^{2r}.$$

Proof. In the ordinary addition of x and x' , let $c_i \in \{0, 1\}$ be the carry entering digit i , with $c_0 = 0$, and let c_{2r} be the final carry. Once u and the vector

$$(c_1, \dots, c_{2r}) \in \{0, 1\}^{2r}$$

are fixed, the i th ordinary output digit is forced to be $(u_i + c_i) \bmod p$, and the final digit is c_{2r} . Thus the resulting integer sum is uniquely determined. Some carry vectors may be inconsistent with any pair of summands, which only reduces the count. $\square$

Define the integer palette

$$B_M = \bigcup_{u \in S} \mathcal{C}(u) \subseteq \{0, \dots, 2M - 2\}.$$

Lemma 6.1 and Proposition 5.1 give

$$|B_M| \leq 2^{2r} p^{r+d}.$$

Let K_{B_M} be the simple graph on $[M]_0$ in which distinct integers x, x' are adjacent when $x + x' \in B_M$. If x, x' form an edge of Γ_S under the digit bijection, then $\psi(x) + \psi(x') \in S$, so the carry-fibre definition (6.1) and the definition of B_M place their ordinary sum in B_M . Therefore K_{B_M} contains the group graph and

$$\alpha(K_{B_M}) \leq p^{r+s}.$$

No carry has been discarded, and only pairs of distinct vertices are used in this graph comparison.

7 Parameters for every interval length

We now choose the field and dimension separately for each N . This also gives the quantitative error term in Theorem 1.1. Let N be sufficiently large, put

$$L = \log N, \quad \lambda_0 = \left(\frac{L}{\log L} \right)^{1/3}, \quad P = \lceil \exp(\lambda_0) \rceil,$$

and choose a prime $p \in [P, 2P]$, which exists by Bertrand's postulate [10]. Set

$$r = \left\lceil \frac{\log N}{2 \log p} \right\rceil, \quad M = p^{2r}. \tag{7.1}$$

Then p is odd and tends to infinity, and

$$\lambda := \log p = \lambda_0 + O(1), \quad r = \Theta(L/\lambda).$$

Since $\log r = O(\log L) = o(\lambda)$, one has $\log(pr) = O(\lambda)$, and (4.8) gives $\ell = O(\lambda)$. Moreover,

$$(\log(pr))^2 \log(2 + \log(pr)) = O(\lambda^2 \log \lambda) = o(L),$$

whereas $r \log p = \Theta(L)$. Thus the hypothesis of Theorem 4.1 holds with room to spare. The design construction and the quantitative bounds in that theorem give

$$\begin{aligned} D &= O\left(\lambda \sqrt{L/\lambda} + \lambda^2 \log \lambda\right) = O\left(L^{2/3} (\log L)^{1/3}\right), \\ d, s &= O\left(L^{1/3} (\log L)^{2/3}\right), \\ \frac{d}{r}, \frac{s}{r} &= O\left(\left(\frac{\log L}{L}\right)^{1/3}\right). \end{aligned}$$

Also, by the ceiling in (7.1),

$$N \leq M < Np^2.$$

Build the M -vertex palette B_M above and restrict to the first N vertices. Delete the normalized sums which cannot come from two distinct members of $[N]_0$ by putting

$$B_N = B_M \cap \{1, \dots, 2N - 3\}. \quad (7.2)$$

Let Γ_N be the subgraph of Γ_S induced by $[N]_0$. Every edge of Γ_N has ordinary sum in $B_M \cap \{1, \dots, 2N - 3\} = B_N$, so

$$\Gamma_N \subseteq H_{B_N}.$$

Therefore

$$\alpha(H_{B_N}) \leq \alpha(\Gamma_N) \leq \alpha(\Gamma_S) < p^{r+s}.$$

Together with the palette-size bound, this gives

$$|B_N| \leq 2^{2r} p^{r+d}, \quad \alpha(H_{B_N}) \leq p^{r+s}.$$

We estimate these expressions relative to N . Put $\lambda = \log p$. By (7.1), $r = L/(2\lambda) + O(1)$. At these parameters the bounds for d and s give $d\lambda, s\lambda = O(D + \log(pr)) = O(D + \lambda)$, so

$$\begin{aligned} \frac{\log(2^{2r} p^{r+d})}{L} &= \frac{2r \log 2}{L} + \frac{(r+d)\lambda}{L} \leq \frac{1}{2} + O\left(\frac{1}{\lambda} + \frac{D}{L} + \frac{\lambda}{L}\right), \\ \frac{\log(p^{r+s})}{L} &\leq \frac{1}{2} + O\left(\frac{D}{L} + \frac{\lambda}{L}\right). \end{aligned}$$

Here the carry factor contributes $2r \log 2/L = (\log 2)/\lambda + O(L^{-1})$, and the ceiling in (7.1) contributes $O(\lambda/L)$. By the choices above,

$$\frac{1}{\lambda}, \frac{D}{L}, \frac{\lambda}{L} = O\left(\left(\frac{\log L}{L}\right)^{1/3}\right).$$

Therefore, for an absolute constant C_1 and all sufficiently large N ,

$$\begin{aligned} |B_N| &\leq N^{\frac{1}{2} + C_1 \left(\frac{\log \log N}{\log N}\right)^{1/3}}, \\ \alpha(H_{B_N}) &\leq N^{\frac{1}{2} + C_1 \left(\frac{\log \log N}{\log N}\right)^{1/3}}. \end{aligned} \quad (7.3)$$

8 Completion of the proof

Take $N = n - 1$ and the normalized palette B_N from (7.2). Select the original sums

$$B = \{2n + 2 + a : a \in B_N\}.$$

Because $1 \leq a \leq 2N - 3 = 2n - 5$, this is a subset of

$$\{2n + 3, \dots, 4n - 3\} \subseteq J_n.$$

The translation $x \mapsto n + 1 + x$ identifies the graph generated by B_N on $[N]_0$ with G_B on I_n . Hence (7.3) gives

$$|B| + \alpha(G_B) \leq n^{\frac{1}{2} + O\left(\left(\frac{\log \log n}{\log n}\right)^{1/3}\right)}.$$

The factor 2 from adding the two bounds contributes only

$$\frac{\log 2}{\log n} = o\left(\left(\frac{\log \log n}{\log n}\right)^{1/3}\right)$$

to the exponent. Replacing the scale $N = n - 1$ by n changes the exponent by a smaller amount. These errors are absorbed in the displayed term. By Proposition 2.1, this is the desired upper bound for $f(n)$. Proposition 3.4 supplies the lower bound, and Theorem 1.1 follows.

For clarity about the quantifiers, the construction above chooses one set B for each n . The preceding estimate for $|B| + \alpha(G_B)$ therefore holds with $\alpha(G_B)$ replaced by $|C|$ for *every* B -admissible C . Since the entire exponent correction displayed above tends to zero, it is smaller than ε for sufficiently large n . This proves the stated assertion for every fixed $\varepsilon > 0$ and every sufficiently large n , not merely along a subsequence.

9 Concluding remarks

The proof separates three interfaces that are easy to conflate. The extractor has p -ary linear outputs but binary code coordinates and a binary seed; the weak-design overlap moment is therefore 2^t , not p^t . Ordering the design sets exposes a further resource: at hybrid row i , the remaining $r - i$ output coordinates pay for the excess above the unavoidable baseline overlap. The union of the resulting kernels controls finite-field sum graphs, and the carry closure in Section 6 then transfers that control to ordinary integer addition. Keeping these interfaces explicit yields the quantitative exponent correction $O((\log \log n / \log n)^{1/3})$.

The argument determines the exponent of $f(n)$ but not its finer order of growth. The universal lower bound is $\Omega(\sqrt{n \log n})$, whereas the constructed upper bound is

$$\sqrt{n} \exp(O((\log n)^{2/3}(\log \log n)^{1/3})).$$

Closing this remaining subpolynomial gap would require a sharper palette, a stronger universal lower bound, or both.

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